



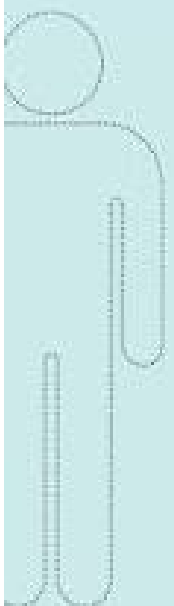
PSYC1003

Memory for Traumatic Events



1 in 4 women

in Australia have
experienced violence
by an intimate partner
since the age of 15.



1 in 14 men

have experienced
violence by an
intimate partner
since the age of 15.



41%

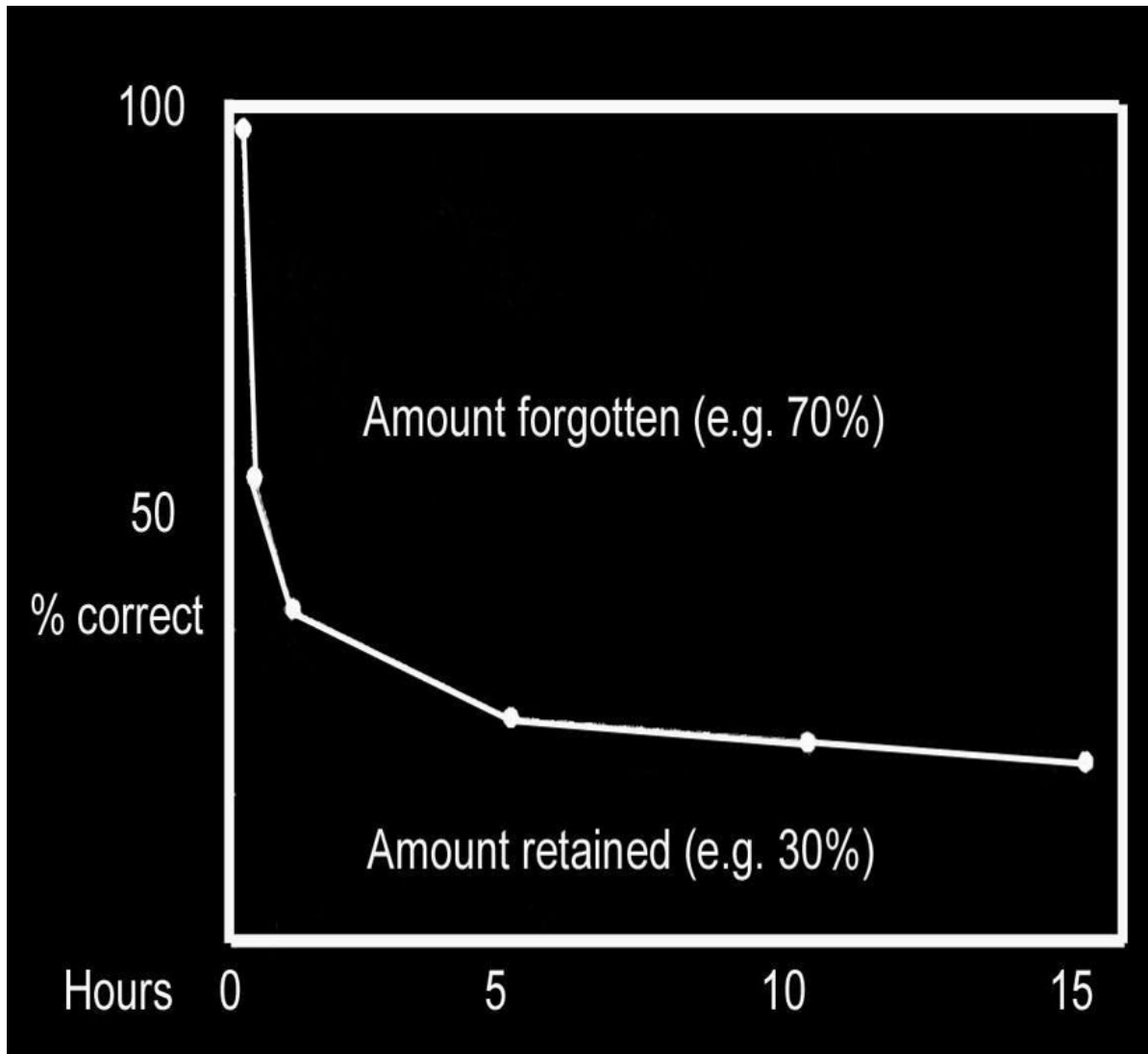
of Australians
(8 million people)
have experienced
physical and/or sexual
violence since age 15.

Learning Objectives

- Explain the reconstructive nature of human memory and its implications for eyewitness testimony and traumatic event recall;
- Describe the physiological, cognitive, and emotional processes involved in emotional arousal and discuss how these processes influence attention and memory;
- Critically evaluate research evidence examining the relationship between stress, emotional arousal, and memory accuracy, including both laboratory and real-world studies;
- Apply the Easterbrook Hypothesis to explain differences in memory for central and peripheral details during emotionally arousing or traumatic events;
- Assess the implications of stress-related memory effects for eyewitness identification, investigative interviewing, and the interpretation of witness testimony.

What can we expect victims to remember?



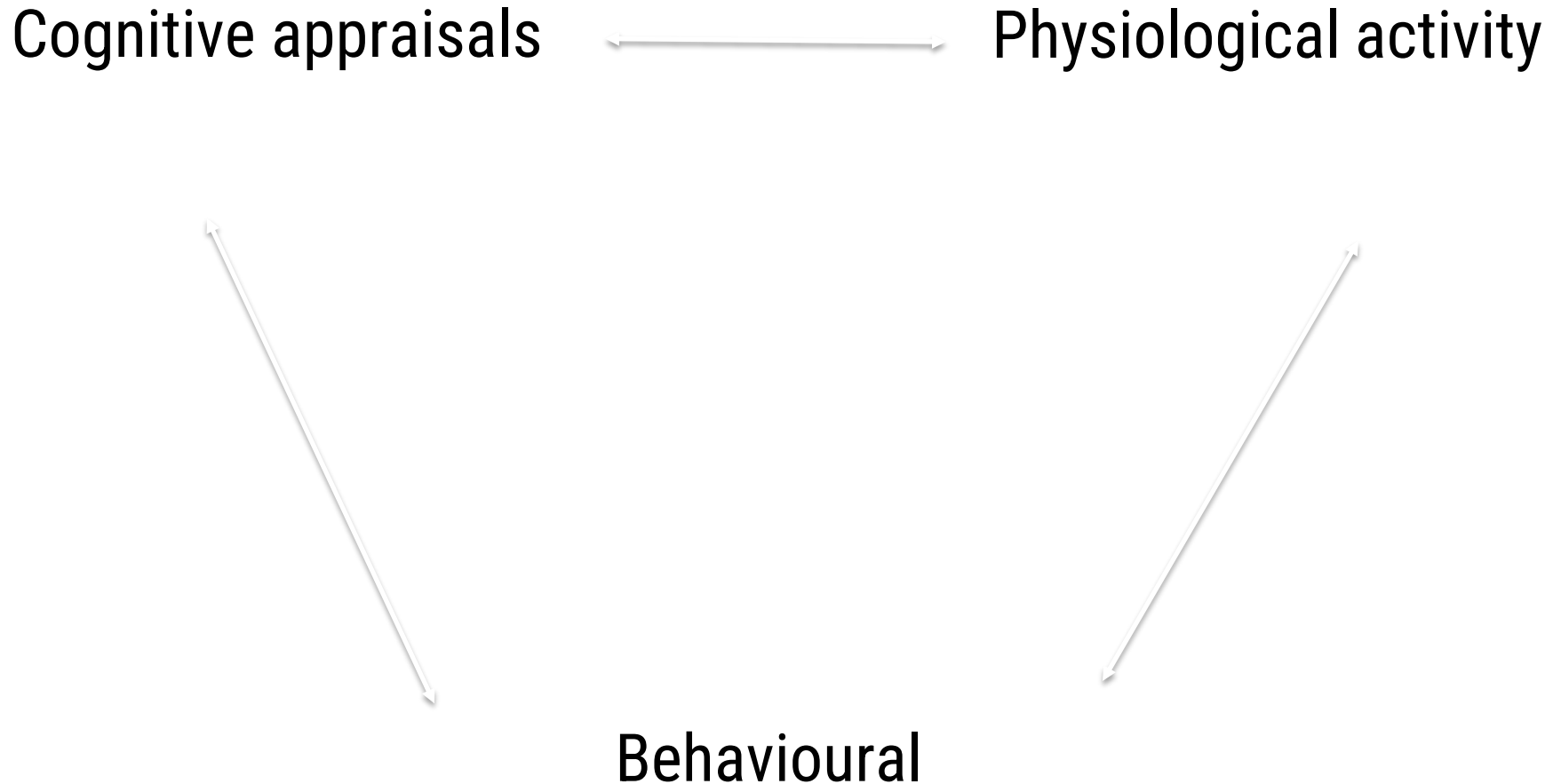


Stress and memory

The three case studies all suggest that stress is detrimental to memory.

However, physiological and psychological research suggests that memory for stressful events should be enhanced.

Emotional Arousal



Cognitive Appraisals

Claimed to be a crucial factor in the experiencing of emotion

Explains why:

- a) Why people experience emotions in some situations but not others.
- b) Individual differences in emotional arousal.

Arousal helps people adapt to their environment

People detect threatening stimuli
faster than neutral ones

Brain wave research shows that
we pay longer attention
to these stimuli

Once attention is drawn, we stay
fixated on emotional stimuli and
this fixation even overrules our
eye blink reflex

In sum:

Threatening stimuli capture our attention instantaneously,
we focus on them, stay fixated on them and are not easily
distracted from them

Why the contradiction between
research and reality?

Research findings discussed so far are largely artificial laboratory- based studies using simple, static, discrete items that are either emotional or neutral.

In contrast, emotionally arousing events that are important for us to remember in real life (e.g., crimes) are likely to be highly complex and dynamic. We need to define qualities (feelings), the situation (e.g. threatening), consequences and bodily reactions

Real Trauma and Memory

13 witnesses of a street shooting were interviewed by police after the incident and by a researcher months later.

Recall accuracy was greatest for those having experienced the greatest degree of stress (trauma) during the crime event.

Past 3 decades = mixed findings:

Early studies suggested that stress impaired eyewitness memory

Later studies suggested that stress enhanced memory

A third body of research suggests that stress can both impair and enhance memory for details of the same emotionally arousing event.

Easterbrook Hypothesis

Highly aroused witnesses have better memory for central details than peripheral details

Weapon Focus Studies

Presence of a weapon draws attention and impairs a witness' ability to identify a culprit

Support for Easterbrook : field studies

- 500+ soldiers, sailors and pilots at 'survival schools' (mock POW camps run by the US military, used to train people to withstand mental & physical stresses of capture).
- After 48 hours without food or sleep, they were subjected to intense interrogation.
 - Half the subjects were physically threatened, causing them to show signs of intense physiological stress.
- 24 hours after release from the camp, the subjects were asked to identify their interrogators.

Participants: 70 Police
Authorised Firearms Officers

Event type: Weapon Present (WP) "shoot" scenario vs. Weapon Absent (WA) "no shoot" scenario.

Participants in the Weapon-Present Condition recalled *fewer* event details than participants in the No-Weapon-Present Condition ...but these details were *more accurate*

Easterbrook's theory supported by a vast number of studies

Memory enhancing effect for central items is expected based on what we know about physiological, evolutionary and psychological functions of emotional arousal.

Memory impairing effect for peripheral details helps us understand apparent memory errors when witnesses try and recall an event and/or identify a perpetrator.

Centrality not arbitrary

Whatever is causing stress, that is where you will focus on. Enhancing memory for those aspects.

The surrounding elements will be peripheral and easily forgotten or distorted.

Stress has an impact on our memories, but understanding those effects can help when asking people to recall these memories.

Learning Outcomes

- Explain the reconstructive nature of memory.
- Describe the effects of emotional arousal on attention and memory.
- Evaluate evidence on the relationship between stress and memory accuracy.
- Apply the Easterbrook Hypothesis to traumatic event memory.
- Assess the implications of stress-related memory effects for eyewitness testimony

Thank you!

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